

# AIMPACT: AI-ENABLED MULTI-MODAL PRESENTATION ASSESSMENT & COACHING TUTOR

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## ABSTRACT

Assessing a recorded presentation against a rubric requires evidence from audio, video, and slides. Expert labels are costly to obtain, experts who apply the same rubric disagree, and general multimodal language models concentrate their scores near the middle of the scale. We present AIMPACT, a multimodal assessment system that scores a presentation in three stages. An evidence-decoupled multi-agent stage converts the three input streams into a shared, traceable evidence record, from which specialist agents score the rubric dimensions. A reliability-guided collaboration graph then estimates the reliability of each dimension’s evidence without labels, activates deliberation only where a disagreeing peer rests on more reliable evidence, and combines the outcome into a council prior. STAR-PO, a rubric-conditioned generative judge built on Qwen3-8B, aligns this prior with expert scoring preferences. It is first fine-tuned on expert corrections to the prior and then optimized on policy under a reward that scores agreement with the confidence-weighted expert distribution together with evidence faithfulness, confidence calibration, and label-free rubric following. On a corpus of 35 classroom presentations annotated by five experts on eight dimensions, the full system reaches a Spearman correlation of 0.833 with the expert consensus, against 0.374 for a transcript-only language-model baseline, and each stage improves the axis it targets. Under matched inputs, AIMPACT ranks presentations more like a human expert than four omni multimodal judges, which compress their scores toward the middle of the scale. Because the rubric is an input, the trained judge transfers to the speech dimensions of SpeechOcean762 without retraining.

## 1 INTRODUCTION

Presentation courses assess students against a rubric and return feedback. The feedback is most useful when it is specific and located in time, for example that speaking pace fell during one section, that the presenter looked at the screen while explaining a diagram, or that a slide was read out without elaboration. Producing feedback at this level of detail requires an instructor to review each recording closely, which is rarely feasible for a class of thirty, so students receive short and general comments, often several days later.

Automating this task requires evidence from three modalities. Delivery is observable in the video, voice quality in the audio, and content is split between what the speaker says and what the slides show. A system has to process all three and combine them. The target is a human judgment produced by applying a rubric, and experienced raters applying the same rubric disagree with each other. A system’s score is useful only to the extent that it agrees with those raters, since feedback derived from a score the instructor would not have assigned carries the same error.

We recorded and annotated a corpus of classroom presentations for this study, described in Section 3. Five expert raters scored each presentation on an eight-dimension rubric, and their ratings shape the method. Raters disagree, so a single aggregated label per presentation discards information about which judgments are settled and which are contested. Raters also use the full scale, while general multimodal language models concentrate their predictions in a narrow band. The four omni judges evaluated here have overall score spreads between 0.34 and 0.41 of the expert’s, and the one with the lowest absolute error still ranks presentations below our system. Deployment adds a further

constraint. Rubrics differ between courses and institutions, and a scoring standard represented in model weights cannot be changed without retraining.

AIMPACT, shown in Figure 1, addresses these constraints in three stages. The first stage decouples evidence from judgment. Frozen perception models and deterministic operators convert the audio, video, and slides into a shared evidence record in which every item keeps its value, location, support, and provenance, and specialist agents score the rubric dimensions from the evidence assigned to them. The second stage builds a collaboration graph over the dimensions. Each dimension’s evidence receives a reliability computed without labels, deliberation is activated only where a disagreeing same-modality peer rests on more reliable evidence, and the outcome is combined into a council prior that carries reliability and unresolved disagreement forward. The third stage is STAR-PO, a rubric-conditioned generative judge that aligns the council prior with expert scoring preferences. It is fine-tuned on expert corrections to the prior and then optimized on policy against the confidence-weighted expert distribution, under a reward that also enforces evidence faithfulness, confidence calibration, and a label-free rubric-following constraint. Because the rubric is an input, the standard can be changed at inference, and we measure this by transferring the judge to the speech dimensions of SpeechOcean762 without retraining.

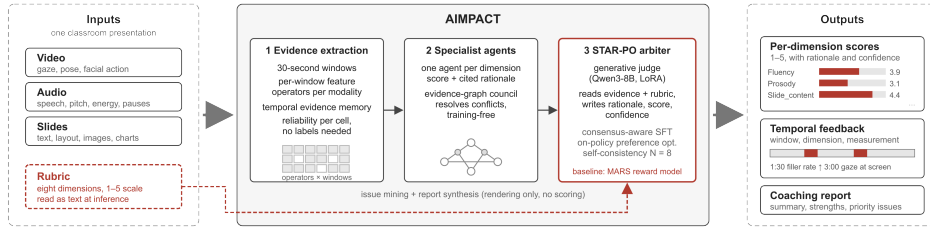


Figure 1: Overview of AIMPACT. It takes in video, audio and slides, then scores the presentation and provides actionable feedback for users.

Our contributions are the following.

- **A classroom presentation corpus and its annotation protocol.** Graded student talks with synchronized video, audio, and slides, scored by five expert raters on an eight-dimension rubric with per-dimension confidence. Every rater’s own judgment is retained. Privacy constraints keep the recordings from being released, discussed in the ethics statement.
- **An evidence-decoupled multi-agent architecture with a reliability-guided collaboration graph.** A shared evidence record separates measurement from judgment, a label-free reliability decides which disagreements deliberation may revise, and the result is a council prior that keeps unresolved disagreement as uncertainty. The same reliability signal predicts when the system is wrong and supports abstention.
- **STAR-PO**, a rubric-conditioned preference-alignment method that trains a generative judge around the council prior, with supervised fine-tuning on expert corrections followed by on-policy preference optimization against the confidence-weighted expert distribution, under label-free rubric-following and paraphrase-consistency constraints.
- **An evaluation** that ablates each stage on the axis it targets, compares the system against four omni multimodal language-model judges under input parity, and measures transfer to an unseen corpus and rubric without retraining.

## 2 RELATED WORK

### 2.1 PRESENTATION ASSESSMENT DATASETS

Automated scoring of public speaking combines prosodic, lexical, and visual features with supervised regressors (Chen et al., 2017). Existing corpora label adjacent constructs, persuasiveness in multimodal opinion video (Park et al., 2014) and audience response to conference talks (Haider & Luz, 2019). SpeechOcean762 provides utterance-level expert scores for pronunciation, fluency, and

prosody (Zhang et al., 2021), and we use it as an unseen transfer target. The corpus of Section 3 is labeled by the instructors who teach and assess against this rubric, and it retains each rater’s score and stated confidence per dimension rather than one aggregated label. The individual ratings supply the panel disagreement the judge is trained to reproduce and let us report the consensus reliability with standard inter-rater statistics (Shrout & Fleiss, 1979; Krippendorff, 2011).

## 2.2 ASSESSMENT METHODS

Reducing prosody, gaze, pose, facial action, and slide content to per-window operators follows work on multimodal representation learning (Baltrušaitis et al., 2019) and slide understanding (Tanaka et al., 2023). We use hand-designed operators over pretrained perception models rather than a learned encoder, since every score must trace back to a named measurement and the corpus is too small to fit an encoder. The scoring stage instead relies on a language-model judge. Such judges are widely used for evaluation (Zheng et al., 2023), with documented position and verbosity bias and poor calibration (Wang et al., 2023), and rubric-conditioned evaluators improve fine-grained scoring (Kim et al., 2024). Agreement for a judge is normally reported as a correlation or an error against a reference. We add the spread of its scores, which separates judges that track the reference from judges that predict near its mean.

Training such a judge against expert ratings draws on preference optimization. Reward modeling from pairwise comparisons follows Bradley & Terry (1952) and underlies learning from human feedback (Ouyang et al., 2022). Group-relative updates remove the value network (Shao et al., 2024), and direct preference optimization trains from fixed preference pairs (Rafailov et al., 2023). These methods optimize a policy against a reward standing in for one aggregated human preference. Our labels are scarce and contested, so the reward is built from a panel’s soft label and disagreement rather than a single target, and the standard-following term needs no labels at all, since it supervises the judge with rubric perturbations whose expected effect is known. The council’s reliability signal also supports selective prediction, where a model abstains on inputs it is likely to get wrong (Geifman & El-Yaniv, 2017).

## 3 DATASET

We drew the corpus from a graded assignment in a university speech and music machine learning course, where students delivered a prepared talk with slides to their class. To hold recording conditions constant across presenters, we captured video on a DJI camera framing the speaker and the projected slides, and audio on a DJI wireless microphone worn by the presenter, which keeps the speech signal usable for prosody and fluency measurement in a room with audience noise. We collected the slide decks in their original format and de-identified all recordings before annotation. The corpus has 35 recordings, and the 34 that carry all three streams are used in the system evaluation.

Five expert raters from a university English communication center annotated the corpus. All five are practicing instructors who teach and assess presentation skill against this rubric, so the labels reflect the standard applied in the course. As Figure 2 shows, each rater scored every presentation on eight dimensions covering language, three vocal dimensions, non-vocal delivery, and three visual dimensions (Appendix Table 7), on a 1–5 Likert scale, together with an overall score and a 1–5 confidence per score. We retain the individual rater scores and do not average them at collection time, which makes the spread across raters available as a signal.

From these ratings we derive, for each (video, dimension), a confidence-weighted panel mean  $\bar{s}$ , a panel disagreement  $\sigma$  (the confidence-weighted standard deviation), the panel mean confidence  $\bar{c}$ , and a soft label  $q$  over the five Likert bins. To build  $q$ , we place each rater’s confidence-normalized weight at its score bin and then widen the distribution in proportion to  $\sigma$ , as Appendix A.1 details. We use this notation throughout Section 4.

Because all alignment results in this work use the five-expert consensus as the reference, the consensus has to be reliable enough to serve as an evaluation target. Appendix Table 6 reports intra-class correlations per dimension (Shrout & Fleiss, 1979). Single-rater ICC(2,1) ranges from 0.38 to 0.72, which shows meaningful subjectivity at the individual level. Averaging over five raters raises ICC(2,k) to 0.75 to 0.93, so the consensus is more stable than any one rater. The raters report a mean confidence of 3.96, with more than 80% of ratings at 4 or higher. The eight dimensions are corre-

### Dataset annotation by 6 human experts

Each human expert gives a score across the 8 dimensions and labels points in the video with issues.

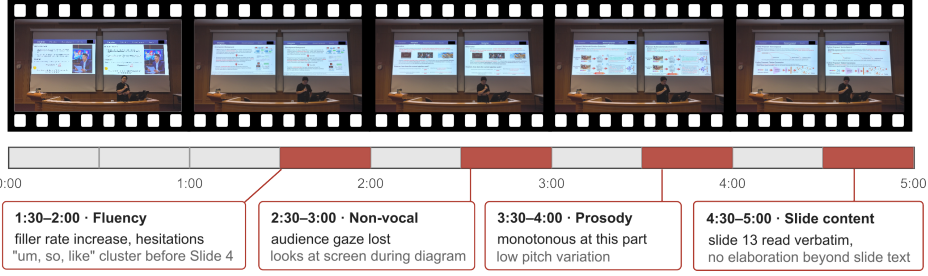


Figure 2: Dataset annotation by human experts who score each presentation and justify the issues.

lated, with the first principal component explaining more than 70% of the variance, which explains why systems can look similar on an aggregate score while differing at the dimension level.

## 4 METHOD

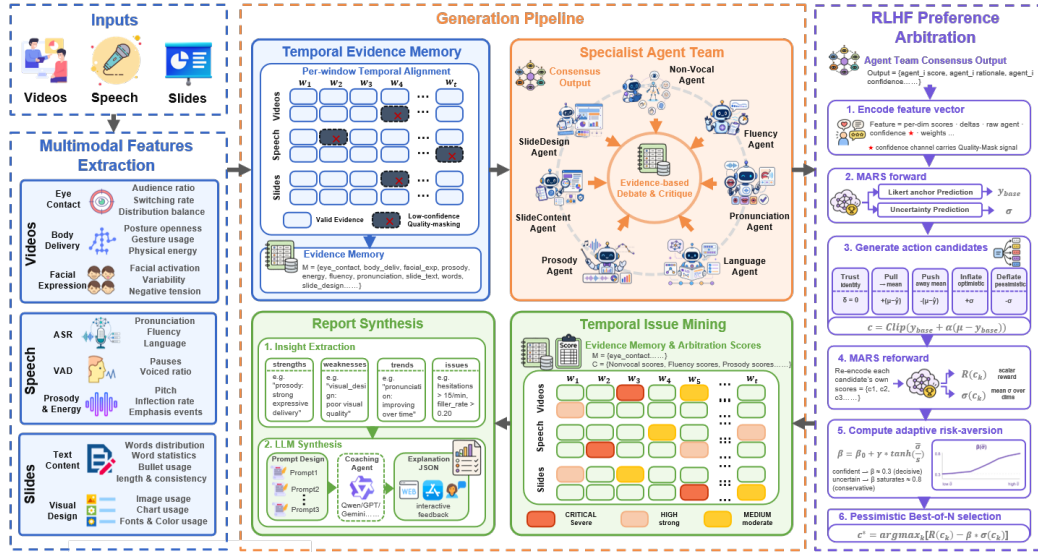


Figure 3: Overview of the AIMPACT framework. Multimodal inputs are first converted into a shared evidence record and evaluated by specialist agents. A reliability-guided collaboration graph then coordinates cross-agent deliberation to form a council prior, which STAR-PO aligns with expert preferences under the corresponding rubric.

### 4.1 ASSESSMENT TAXONOMY

Given synchronized audio, video, and slides, AIMPACT evaluates a presentation across eight scoring dimensions defined by expert instructors, producing a score, rationale, and predicted confidence for each dimension. Let  $d$  index a scoring dimension and  $\mathcal{R}_d$  denote its corresponding rubric. As shown in Figure 3, the assessment goes through three stages comprising multi-agent assessment, graph collaboration, and human preference alignment. The first stage constructs the evidence set  $E_d$  required by each dimension, from which a specialist agent produces the score  $s_d$ . Building on these scores, the second stage estimates the evidence reliability  $\rho_d$  to regulate cross-agent deliberation, producing the revised score  $s'_d$ , retaining unresolved disagreement as uncertainty  $u_d$ , and forming the council prior  $\tilde{s}_d$ . Finally, the third stage uses the expert scores  $s_{rd}$  to align  $\tilde{s}_d$  with human scor-

ing preferences under rubric  $\mathcal{R}_d$ , yielding the final score  $\hat{s}_d$  together with its rationale and predicted confidence. The following sections detail the aforementioned three stages, respectively.

## 4.2 EVIDENCE-DECOUPLED MULTI-AGENT ASSESSMENT

Directly mapping heterogeneous multimodal signals to rubric-based scores entangles measurement noise with scoring decisions, making it difficult to distinguish weak evidence from incorrect assessment. To address this problem, we introduce an evidence-decoupled multi-agent framework organized as a hierarchical assessment architecture. At the evidence layer, synchronized audio, video, and slides are processed by frozen pretrained models for word-aligned speech recognition (Radford et al., 2022), gaze estimation (Abdelrahman et al., 2023; Kellnhofer et al., 2019), pose tracking (Jiang et al., 2023), and facial action unit detection (Chang et al., 2024), together with deterministic multimodal operators that measure speech timing, pronunciation confidence, prosodic variation, slide design, and speech-slide alignment. Their outputs are organized into a shared evidence record in which each evidence item preserves its observed value, temporal or slide location, number of valid supporting observations, operator confidence, and source provenance, providing a common and traceable basis that remains independent of subsequent scoring. Built on this evidence layer, seven specialist agents evaluate fluency, prosody, pronunciation, language, non-vocal delivery, visual design, and slide content using only the evidence assigned to their respective dimensions. The slide content agent is evaluated under separate synchronization and slide-reference rubrics, allowing the seven agents to cover all eight scoring dimensions. Each agent is constrained by its evidence access and output schema, producing the specialist score  $s_d$  together with a rationale linked to its evidence set  $E_d$ . This hierarchical design separates multimodal evidence construction from rubric-based assessment while maintaining a shared evidence basis across specialist agents. Detailed agent calibration, prompts, output schemas, and multimodal operator settings are reported in Appendix X.

## 4.3 COLLABORATION GRAPH

Although the separation between evidence and specialist layers can prevent evidence from leaking across dimensions, the restricted views of individual agents introduce view-dependent bias and cross-agent disagreement. To address this limitation, we propose a reliability-guided collaboration graph over the specialist layer to determine which scores warrant further deliberation and which disagreements should remain unchanged. Specifically, we construct a directed graph  $\mathcal{G} = (\mathcal{V}, \mathcal{E})$  over the scoring dimensions, where each vertex  $v_d \in \mathcal{V}$  retains the specialist score  $s_d$  and its evidence set  $E_d$ . Before activating cross-agent deliberation, we estimate the reliability  $\rho_d$  of the evidence supporting dimension  $d$  as:

$$\rho_d = \frac{1}{|E_d|} \sum_{e \in E_d} c_e \left( 1 - \exp \left( -\frac{n_e}{\tau_e} \right) \right), \quad (1)$$

where  $n_e$  denotes the number of valid observations supporting evidence  $e$ ,  $c_e$  is the confidence reported by its multimodal operator, and  $\tau_e$  is a scale parameter controlling how rapidly additional observations increase reliability. Missing evidence contributes zero reliability, while  $c_e = 1$  when no confidence estimate is available. Evidence below the grounding threshold is removed before the specialist score  $s_d$  is re-evaluated, allowing evidence quality to govern subsequent collaboration rather than rescaling the specialist score after inference.

The specialist scores  $s_d$  and evidence reliabilities  $\rho_d$  jointly determine the collaboration topology. Let  $\mathcal{N}(d)$  denote candidate dimensions that share the same modality as  $d$ , and let  $\mathbf{A}$  denote the adjacency matrix of the collaboration graph. For each  $j \in \mathcal{N}(d)$ , an edge is activated only when both the score discrepancy and reliability margin are sufficiently large:

$$A_{jd} = \begin{cases} 1, & |s_j - s_d| \geq \tau_s \text{ and } \rho_j - \rho_d \geq \tau_\rho, \\ 0, & \text{otherwise,} \end{cases} \quad (2)$$

where  $\tau_s$  denotes the minimum score discrepancy and  $\tau_\rho$  is the minimum reliability margin required for collaboration. When  $A_{jd} = 1$ , agents  $j$  and  $d$  enter a constrained deliberation in which  $s_j$  serves as the reference for revising  $s_d$  because it is supported by more reliable evidence, using only the scores, rationales, and evidence already available to the connected agents. When  $A_{jd} = 0$ , no deliberation is initiated for that pair. Any disagreement that persists after deliberation is retained as uncertainty  $u_d$ , while the revised score is denoted by  $s'_d$ .

To form the score passed to human preference alignment, the revised score  $s'_d$  is combined with the mean specialist score  $\bar{s}$  retained before deliberation as a fixed global reference. The council prior is defined as:

$$\tilde{s}_d = \gamma_d s'_d + (1 - \gamma_d) \bar{s}, \quad (3)$$

where  $\gamma_d \in [0, 1]$  is a fixed dimension-specific coefficient controlling the contribution of the revised score relative to the global reference. In contrast to the presentation-dependent evidence reliability  $\rho_d$ ,  $\gamma_d$  captures systematic differences in the scoring route across dimensions. The collaboration graph therefore outputs  $(E_d, \tilde{s}_d, \rho_d, u_d)$  for the subsequent human preference alignment stage. Further details of evidence grounding, constrained deliberation, uncertainty estimation, and the dimension-specific coefficients are provided in Appendix X.

#### 4.4 STAR-PO: RUBRIC-CONDITIONED PREFERENCE ALIGNMENT

The council prior  $\tilde{s}_d$  summarizes the evidence-based assessment, but this prior does not yet capture how expert raters apply rubric  $\mathcal{R}_d$ . To align the evidence-based assessment with human scoring preferences without relearning the score from scratch, we introduce STAR-PO and train it in two stages. For dimension  $d$ , the policy receives  $x_d = (E_d, \tilde{s}_d, \rho_d, u_d, \mathcal{R}_d)$  and predicts a score distribution together with a rationale and predicted confidence. The first stage uses supervised fine-tuning (Ouyang et al., 2022) to learn expert corrections around the council prior. Given the score  $s_{rd}$  assigned by expert rater  $r$ , we define the supervised target as

$$s_{rd}^* = \tilde{s}_d + \omega (s_{rd} - \tilde{s}_d), \quad (4)$$

where  $s_{rd} - \tilde{s}_d$  is the expert correction to the council prior and  $\omega \in [0, 1]$  controls its contribution to the target. This initialization learns expert corrections without discarding the council prior, and the resulting SFT policy is frozen as  $\pi_{\text{ref}}$  for preference optimization.

The second stage performs on-policy preference optimization using the full expert panel. Since experts may assign different scores under the same rubric, we form a confidence-weighted expert distribution  $q_d$  instead of averaging their scores, thereby retaining disagreement as part of the training signal. Starting from  $\pi_{\text{ref}}$ , the current policy  $\pi_\theta$  samples assessments  $o$  and optimizes the KL-regularized objective (Ouyang et al., 2022; Kirk et al., 2024):

$$\mathcal{J}(\theta) = \mathbb{E}_{x_d \sim \mathcal{D}} [\mathbb{E}_{o \sim \pi_\theta(\cdot | x_d)} R_{\text{STAR}}(o; x_d, q_d) - \beta_{\text{KL}} D_{\text{KL}}(\pi_\theta(\cdot | x_d) \| \pi_{\text{ref}}(\cdot | x_d))], \quad (5)$$

where  $\mathcal{D}$  denotes the training set and  $\beta_{\text{KL}}$  controls the regularization toward  $\pi_{\text{ref}}$ . The reward  $R_{\text{STAR}}$  uses agreement with  $q_d$  as its primary preference signal and adds terms for evidence faithfulness, confidence calibration, and rubric following. For each  $x_d$ , samples from the current policy are optimized using signed group-relative advantages (Shao et al., 2024). The KL regularization constrains this update around the SFT reference policy, while an additional diversity term in  $R_{\text{STAR}}$  discourages excessive concentration around middle-range scores.

To ensure that the rubric directly governs scoring,  $R_{\text{STAR}}$  includes a rubric-following term based on controlled changes to  $\mathcal{R}_d$ . Holding the remaining input fixed, we replace  $\mathcal{R}_d$  with stricter and looser variants to obtain  $x_d^-$  and  $x_d^+$ , and constrain the corresponding predictions as:

$$\hat{s}_\theta(x_d^-) + \delta \leq \hat{s}_\theta(x_d) \leq \hat{s}_\theta(x_d^+) - \delta, \quad (6)$$

where  $\delta > 0$  specifies the minimum score change required when the rubric changes. We further require meaning-preserving paraphrases of  $\mathcal{R}_d$  to yield consistent predictions, so that score variation reflects changes in the rubric itself rather than differences in wording. At inference, multiple samples are averaged to obtain the final score  $\hat{s}_d$ , while the associated rationale and predicted confidence are retained for feedback. Additional details on the expert distribution, reward formulation, model adaptation, and optimization are provided in Appendix X.

## 5 EXPERIMENTS

### 5.1 EXPERIMENTAL SETUP

**Implementation Details.** The reference for alignment is the five-expert confidence-weighted consensus of Section 3. All learned components are evaluated under five-fold cross-validation with folds split over videos, and the fold assignment is frozen across every configuration so that all systems see identical held-out videos. The judge is trained per fold and scored on its held-out fold only.

Table 1: Agreement on the main dataset under matched multimodal inputs. Models are grouped by deployment class and ordered by parameter count within each group, with consensus correlation breaking size ties. CI denotes the 95% bootstrap confidence interval of Spearman’s  $\rho$ , and VR is  $\sigma_j/\sigma_h$ . Best results are bold and second-best results are underlined.

| Method                                            | Single expert   |               |                  |                   |                    | Five-expert consensus |               |                  |                   |                    |
|---------------------------------------------------|-----------------|---------------|------------------|-------------------|--------------------|-----------------------|---------------|------------------|-------------------|--------------------|
|                                                   | $\rho \uparrow$ | 95% CI        | MAE $\downarrow$ | $\alpha \uparrow$ | VR $\rightarrow 1$ | $\rho \uparrow$       | 95% CI        | MAE $\downarrow$ | $\alpha \uparrow$ | VR $\rightarrow 1$ |
| <i>Proprietary model</i>                          |                 |               |                  |                   |                    |                       |               |                  |                   |                    |
| Gemini 3.1 Pro                                    | <u>0.599</u>    | [0.36, 0.78]  | <u>0.62</u>      | <b>0.56</b>       | 0.36               | <u>0.764</u>          | [0.56, 0.89]  | <u>0.39</u>      | <b>0.58</b>       | 0.55               |
| <i>Large MoE models (30B total; 3B activated)</i> |                 |               |                  |                   |                    |                       |               |                  |                   |                    |
| Qwen3-Omni-30B-A3B-T                              | 0.228           | [-0.14, 0.56] | 0.76             | 0.28              | <b>0.64</b>        | 0.424                 | [0.10, 0.68]  | 0.55             | 0.49              | <b>1.00</b>        |
| Qwen3-Omni-30B-A3B-I                              | 0.173           | [-0.17, 0.49] | 0.89             | -0.18             | 0.41               | 0.323                 | [-0.00, 0.60] | 0.59             | -0.26             | 0.64               |
| <i>Comparable open-weight models (7–9B)</i>       |                 |               |                  |                   |                    |                       |               |                  |                   |                    |
| Ola-7B                                            | 0.131           | [-0.19, 0.43] | 0.75             | 0.12              | <u>0.53</u>        | 0.295                 | [-0.00, 0.55] | 0.59             | 0.15              | <u>0.82</u>        |
| Baichuan-Omni-1.5 (7B)                            | 0.085           | [-0.26, 0.42] | 0.93             | -0.15             | 0.37               | 0.281                 | [-0.09, 0.60] | 0.63             | -0.34             | 0.57               |
| Qwen2.5-Omni-7B                                   | -0.007          | [-0.35, 0.35] | 1.15             | -0.24             | 0.34               | 0.073                 | [-0.28, 0.43] | 0.91             | -0.37             | 0.53               |
| MiniCPM-o-2.6 (8B)                                | 0.215           | [-0.07, 0.46] | 0.71             | 0.18              | 0.47               | 0.147                 | [-0.20, 0.45] | 0.65             | 0.12              | 0.73               |
| MiniCPM-o-4.5 (9B)                                | 0.493           | [0.15, 0.74]  | 0.90             | 0.03              | 0.41               | 0.447                 | [0.08, 0.72]  | 0.61             | -0.09             | 0.63               |
| <b>AIMPACT (8B, ours)</b>                         | <b>0.791</b>    | [0.59, 0.90]  | <b>0.57</b>      | <u>0.53</u>       | 0.48               | <b>0.833</b>          | [0.68, 0.91]  | <b>0.30</b>      | <u>0.54</u>       | 0.69               |

**Metrics and Baseline.** Each component is measured on the axis it is intended to improve. Alignment with the consensus is evaluated with Spearman’s  $\rho$ , mean absolute error, ordinal Krippendorff’s  $\alpha$  (Krippendorff, 2011), and the variance ratio  $\sigma_{\text{judge}}/\sigma_{\text{human}}$ , where values below 1.0 indicate central-tendency collapse. Reliability is evaluated with the AUROC of the label-free reliability signal for detecting scoring errors. Transfer is evaluated on 500 utterances of SpeechOcean762 (Zhang et al., 2021) with no retraining, by swapping in that corpus’s scoring standard as the judge’s rubric input. That corpus carries audio only and covers pronunciation, fluency, and speaking rate, so we score it with Pearson, Spearman, and Kendall correlations. The remaining metrics assume that the model output and the reference share an absolute scale, which does not hold once the rubric is replaced. Confidence intervals are 95% percentile-bootstrap intervals from 10,000 resamples at the video or utterance level with a fixed seed.

The ablation of Section 5.3 starts from a transcript-only baseline, where a language model predicts the eight dimension scores directly from the raw transcript with no evidence extraction. The external comparison of Section 5.2 places the system against multimodal language-model judges under input parity. Every judge receives the video frames, audio, transcript, and slide images for the same presentation, so a difference in agreement reflects how the judge uses that evidence and not what it was allowed to see. This restricts the baseline set to models that accept audio and video. We use seven open, locally served models, Qwen2.5-Omni-7B, Qwen3-Omni-30B-A3B-Instruct, Qwen3-Omni-30B-A3B-Thinking, MiniCPM-o-2.6, MiniCPM-o-4.5, Ola-7B, and Baichuan-Omni-1.5, together with one closed model, Gemini 3.1 Pro, accessed through a paid API. We score this comparison against both references, a single expert annotator and the five-expert consensus, and dimension-wise comparisons use paired bootstrap tests with Holm-Bonferroni correction. On SpeechOcean762 the baseline set is further restricted to the models that accept audio-only input.

## 5.2 COMPARISON ANALYSIS

Table 1 reports agreement on the main dataset under matched inputs. AIMPACT reaches a Spearman correlation of 0.791 against single expert and 0.833 against five-expert consensus, the highest under both references. The closest baseline is Gemini 3.1 Pro at 0.599 and 0.764. AIMPACT also records the lowest mean absolute error under both references, 0.57 and 0.30. The remaining seven baselines stay below 0.45 against the consensus, and four of them have confidence intervals that include zero.

The scale-matched comparison separates the gain from backbone size. Among the 7–9B baselines, MiniCPM-o-4.5 is strongest by consensus correlation, yet the 8B AIMPACT improves  $\rho$  by 0.386 and reduces MAE by 0.31. Increasing the backbone to the 30B-A3B Qwen3-Omni variants does not close this gap: the stronger thinking variant reaches  $\rho = 0.424$ . Its variance ratio of 1.00 shows that model scale can recover the human score range without recovering the human ranking. The advantage of AIMPACT therefore comes from the assessment pipeline rather than a larger underlying model.

Table 2: Out-of-domain and reliability evaluation. (a) Zero-shot transfer to 500 SpeechOcean762 utterances. (b) Association between confidence and negative prediction error. CI is the 95% bootstrap interval for Spearman’s  $\rho$ .

| (a) Cross-domain transfer |              |                 |               |                 | (b) Confidence–correctness                    |                 |                   |
|---------------------------|--------------|-----------------|---------------|-----------------|-----------------------------------------------|-----------------|-------------------|
| Method                    | $r \uparrow$ | $\rho \uparrow$ | 95% CI        | $\tau \uparrow$ | Confidence signal                             | $\rho \uparrow$ | $p$               |
| Qwen2.5-Omni-7B           | <u>0.197</u> | 0.179           | [0.12, 0.23]  | 0.161           | Naive LLM self-consistency                    | <u>0.079</u>    | 0.002             |
| Baichuan-Omni-1.5 (7B)    | 0.022        | 0.007           | [−0.05, 0.06] | 0.007           | Naive LLM verbalized conf.                    | −0.003          | 0.902             |
| Ola-7B                    | 0.000        | 0.000           | [0.00, 0.00]  | 0.000           | Audio-LLM softmax                             | −0.000          | 0.997             |
| MiniCPM-o-2.6 (8B)        | 0.184        | <u>0.185</u>    | [0.13, 0.24]  | <u>0.165</u>    |                                               |                 |                   |
| <b>AIMPACT (8B, ours)</b> | <b>0.258</b> | <b>0.274</b>    | [0.23, 0.32]  | <b>0.204</b>    | <b>Reliability <math>\rho_d</math> (ours)</b> | <b>0.107</b>    | <b>&lt; 0.001</b> |

Table 3: Cumulative component ablation against the five-expert consensus. Check marks indicate enabled components; Err-AUC evaluates error detection and  $\rho_{\text{OOD}}$  zero-shot transfer. Entries marked  $\approx$  are estimates.

| Configuration               | Enabled components |      |        |         |                    |                  | Alignment       |                  | Reliability        | Transfer                     |
|-----------------------------|--------------------|------|--------|---------|--------------------|------------------|-----------------|------------------|--------------------|------------------------------|
|                             | Agents             | Gate | Delib. | STAR-PO | $A_{\text{align}}$ | $A_{\text{std}}$ | $\rho \uparrow$ | MAE $\downarrow$ | Err-AUC $\uparrow$ | $\rho_{\text{OOD}} \uparrow$ |
| Transcript-only LLM         | –                  | –    | –      | –       | –                  | –                | 0.374           | 0.86             | 0.499              | −0.018                       |
| Multimodal specialists      | ✓                  | –    | –      | –       | –                  | –                | 0.633           | 0.62             | 0.499              | 0.294                        |
| Reliability-gated evidence  | ✓                  | ✓    | –      | –       | –                  | –                | 0.704           | 0.53             | 0.548              | $\approx$ 0.29               |
| Conflict-graph deliberation | ✓                  | ✓    | ✓      | –       | –                  | –                | 0.722           | 0.38             | 0.548              | 0.306                        |
| Fixed regressor aggregation | ✓                  | ✓    | ✓      | –       | –                  | –                | 0.623           | 0.32             | 0.548              | $\approx$ 0.29               |
| STAR-PO, accuracy only      | ✓                  | ✓    | ✓      | ✓       | –                  | –                | 0.816           | 0.32             | 0.548              | $\approx$ 0.37               |
| + $A_{\text{align}}$        | ✓                  | ✓    | ✓      | ✓       | ✓                  | –                | 0.801           | 0.33             | 0.548              | $\approx$ 0.38               |
| <b>Full AIMPACT</b>         | ✓                  | ✓    | ✓      | ✓       | ✓                  | ✓                | <b>0.833</b>    | <b>0.30</b>      | 0.548              | <b>0.274</b>                 |

Correlations rise for most systems when the reference changes from one expert to the panel, which follows from averaging five raters and removing part of the individual rater noise. Two systems move in the other direction. MiniCPM-o-4.5 falls from 0.493 to 0.447 and MiniCPM-o-2.6 from 0.215 to 0.147, so their agreement with the single expert does not carry over to the panel.

The two remaining metrics order the systems differently from the correlations. Gemini 3.1 Pro records the higher ordinal  $\alpha$  under both references, 0.56 against our 0.53 and 0.58 against our 0.54, while ranking presentations less closely under both. On the variance ratio, Qwen3-Omni-30B-Thinking comes closest to the human spread, 0.64 against the single expert and 1.00 against the consensus, with a consensus correlation of 0.424. Using the full width of the scale is therefore consistent with a middling ranking, and we read the ratio alongside the correlations rather than in place of them. Every other system stays below 1.0 under both references, which shows that concentration toward the middle of the scale is common across this baseline set.

Table 2 repeats the comparison on SpeechOcean762, which tests whether the judge applies a rubric it was not trained on. AIMPACT leads on all three correlations, and its Spearman interval is largely separated from those of the two strongest baselines. The two systems built mainly around video input degrade on fine-grained audio-only scoring. Ola-7B assigns the same score to every utterance and Baichuan-Omni-1.5 assigns the same score to more than 90% of them, which leaves both near zero correlation. The absolute values are modest for every system, which reflects how little of the rubric this read-speech corpus can support.

### 5.3 ABLATION STUDIES

Table 3 builds the full system incrementally from the transcript-only baseline. The configurations are cumulative from top to bottom, each row adds one component used in the full method, and the final row is the complete system.

The multimodal evidence and specialist agents carry the largest single gain in alignment, from 0.374 to 0.633, and they also give the judge something to transfer, since the transcript-only baseline is at chance on SpeechOcean762 while evidence-based configurations reach 0.29 and above. Reliability gating raises alignment to 0.704 and is the step that makes the system’s confidence informative,



Table 4: Calibration to the expert-score distribution ( $\downarrow$ );  $p$  compares accuracy only with the full recipe.

| Metric                                   | Accuracy only | + $A_{\text{align}}$ | Full         | $p$          |
|------------------------------------------|---------------|----------------------|--------------|--------------|
| $D_{\text{KL}}(q \parallel p)$           | 2.109         | <u>1.994</u>         | <b>1.536</b> | 0.130        |
| Brier score                              | 0.460         | <u>0.425</u>         | <b>0.350</b> | <b>0.031</b> |
| Total variation                          | 0.520         | <u>0.503</u>         | <b>0.449</b> | 0.074        |
| Spread gap $ \text{sd}_p - \text{sd}_q $ | 0.433         | <u>0.418</u>         | <b>0.352</b> | 0.179        |
| Mean abs. error                          | 0.661         | <u>0.636</u>         | <b>0.584</b> | 0.296        |

Table 5: Retained-set Spearman  $\rho$  by coverage ( $\uparrow$ ); AUC summarizes the curve.

| Signal                                        | Coverage     |              |              | AUC $\uparrow$ |
|-----------------------------------------------|--------------|--------------|--------------|----------------|
|                                               | 100%         | 75%          | 50%          |                |
| LLM verbalized conf.                          | 0.253        | <u>0.231</u> | <u>0.237</u> | 0.224          |
| Audio-LLM softmax                             | 0.197        | 0.224        | <u>0.237</u> | <u>0.234</u>   |
| <b>Reliability <math>\rho_d</math> (ours)</b> | <b>0.289</b> | <b>0.348</b> | <b>0.404</b> | <b>0.358</b>   |

moving error-detection AUROC from chance to 0.548 (Section 5.4.2). Deliberation adds a further gain in  $\rho$  and a large drop in MAE, from 0.53 to 0.38. Within the judging stage, the fixed-regressor control falls back to 0.623, so a learned aggregator over the same evidence is not enough. The STAR-PO judge with accuracy rewards alone reaches 0.816. Adding  $A_{\text{align}}$  costs 1.5 points of  $\rho$ , and Section 5.4.1 shows what it buys in calibration. Adding  $A_{\text{std}}$  completes the reward and gives the best alignment figures,  $\rho = 0.833$  and MAE 0.30, together with the transfer correlation of 0.274 reported in Table 2. Alignment and reliability each improve at the step designed to improve them, and transfer is carried by the evidence stage rather than by any single reward term.

## 5.4 ADDITIONAL ANALYSES

### 5.4.1 EFFECT OF $A_{\text{ALIGN}}$ ON CALIBRATION

The main ablation shows a small loss in pointwise alignment when  $A_{\text{align}}$  is added, so we assess its value on the target objective, the match between the judge’s score distribution and the empirical distribution of expert ratings. For each fold we draw 32 samples per held-out example and normalize the scores into a predicted distribution. Table 4 compares the three learned-judge variants of Table 3.

All five metrics improve when  $A_{\text{align}}$  is introduced, and the full recipe improves every metric further. The Brier score, which needs no smoothing, reaches significance at  $p = 0.031$ . The improvement comes mainly from correcting the most poorly calibrated cases. This term trades under two points of pointwise alignment for a consistent gain in calibration, which is its intended role.

### 5.4.2 RELIABILITY AS A SELECTIVE-PREDICTION SIGNAL

The dimension-level reliability  $\rho_d$  of Section 4.3 is computed without labels, so a natural question is whether it predicts when the system is wrong. Rank correlation between confidence signals and negative error gives  $\rho_d$  the strongest relationship,  $+0.107$  at  $p < 0.001$ , against  $+0.079$  for naive-LLM self-consistency,  $-0.003$  for verbalized confidence, and  $-0.000$  for an external audio-LLM’s softmax. Table 5 evaluates the same signals for abstention.  $\rho_d$  is the only signal whose retained-set correlation improves monotonically as coverage falls, from 0.289 at full coverage to 0.404 at 50%. The two baseline signals stay flat or deteriorate as uncertain samples are removed. The system can abstain on uncertain cases and hand them to an instructor.

## 6 CONCLUSION

We presented AIMPACT, a multimodal presentation-assessment system built from specialist evidence agents, an evidence graph with label-free reliability, and STAR-PO, a generative judge trained with consensus-aware supervised fine-tuning and on-policy preference optimization under a task-specific reward. On a corpus of classroom presentations annotated by five experts, the full system reaches a Spearman correlation of 0.833 with the consensus, with complementary gains from the evidence pipeline, the reliability graph, and the judge. Under matched inputs, four omni multimodal judges compress their scores toward the middle of the scale, and the system stays near the expert’s spread. Because the scoring standard is an input, the judge transfers to the speech dimensions of SpeechOcean762 without retraining. A larger benchmark would tighten these intervals and let transfer be measured on rubrics beyond speech.

## AI USE STATEMENT

In this work, we used generative AI tools for editing and proofreading manuscript text, and for assisting with code review and debugging, under author direction. We have not used generative AI tools for generating research ideas, designing experiments, or producing any experimental results or claims independent of author verification, and literature discovery/summarization and automated plagiarism or novelty checking are not applicable to this work. We have reviewed all AI-assisted work. All experiments were designed, executed, and their results verified by the authors prior to any AI involvement. AI-assisted edits to code were tested against the underlying JSON result files to confirm numerical consistency, and all manuscript text produced or modified with AI assistance was read and edited by the authors for accuracy and correctness of scientific claims. We take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI.

## REPRODUCIBILITY STATEMENT

The recordings cannot be released. We make available the code, the configurations, and the per-dimension rubric text, and results on this corpus are not externally reproducible. The RLHF package builds the judge’s dataset with `build_dataset.py`, trains the two stages under `--stage sft` and then `--stage po --grpo`, and evaluates with `evaluate_starpo.py`. The panel consensus is constructed in `panel_dataset.py` and the reward in `reward.py`. The five folds are frozen so every configuration sees identical held-out videos. The deployed hyperparameters are  $K = 6$ , sampling temperature 1.0,  $\beta = 0.15$ ,  $\lambda_{\text{cov}} = 0.5$ , learning rate  $10^{-4}$ , two epochs, and 4-bit LoRA on Qwen3-8B.

## ETHICS STATEMENT

The corpus records identifiable students in a graded setting. The power imbalance between a course and its participants requires care in obtaining consent. Presenters consented to research use of their recordings, and the material was de-identified before annotation. The recordings contain faces, voices, and slide content belonging to identifiable students. Consent covers research use and not publication, so the corpus is not released and remains access-controlled within the institution that collected it. This limits external replication of the results reported here. Expert ratings are subjective and can encode reviewer preferences. This is one reason we train on five raters and model their disagreement. The intended use is assistive coaching that gives a student timed, evidence-linked feedback. The system is not intended for high-stakes gatekeeping, and its scores are one input to a human decision.

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Table 6: Annotation quality per dimension, computed from the 5-rater by 35-presentation rating matrix. ICC(2,1) is the reliability of a single rater, ICC(2,k) the reliability of the five-rater mean, and  $\bar{r}$  the mean pairwise correlation among raters.

| Dimension          | ICC(2,1) | ICC(2,k) | $\bar{r}$ |
|--------------------|----------|----------|-----------|
| Non-vocal delivery | 0.717    | 0.927    | 0.794     |
| Volume & pace      | 0.613    | 0.888    | 0.678     |
| Fluency            | 0.577    | 0.872    | 0.625     |
| Language           | 0.504    | 0.836    | 0.573     |
| Pronunciation      | 0.479    | 0.821    | 0.573     |
| Speech–slide sync  | 0.443    | 0.799    | 0.553     |
| Slide design       | 0.380    | 0.754    | 0.396     |
| Slide reference    | 0.377    | 0.752    | 0.524     |

## A APPENDIX

### A.1 PANEL CONSENSUS AND SOFT LABEL

For one (video, dimension) with rater scores  $s_r$  and confidences  $c_r$ , let  $w_r = c_r / \sum_{r'} c_{r'}$ . The panel quantities are

$$\bar{s} = \sum_r w_r s_r, \quad \sigma = \left( \sum_r w_r (s_r - \bar{s})^2 \right)^{1/2}, \quad q \propto \left( \sum_r w_r e_{s_r} \right) * [\alpha, 1, \alpha], \quad \alpha = \min(0.5, 0.6\sigma), \quad (7)$$

where  $e_s$  is the one-hot vector at bin  $s$ ,  $*$  is convolution over the five bins, and  $q$  is renormalized to sum to one.  $\bar{c}$  is the unweighted mean of the  $c_r$ . When  $\sigma = 0$  the soft label is the confidence-weighted histogram itself.

### A.2 ANNOTATION QUALITY

Table 6 reports the inter-rater statistics summarized in Section 3.

### A.3 STAR-PO REWARD TERMS IN FULL

The reward is computed per candidate by `reward.py`.  $A_{\text{acc}} = q[\hat{y}] \in [0, 1]$  is the panel soft-label mass at the sampled integer score.  $A_{\text{reg}} = -|\mathbb{E}[p_\theta] - \bar{s}|/4 \in [-1, 0]$  is the negative absolute difference between the model’s expected score and the panel mean, divided by four.  $A_{\text{align}} = D_{\text{KL}}(q \| p_\theta) \geq 0$  is the KL divergence from the panel soft label to the model distribution.  $A_{\text{faith}} = 0.5g + 0.5h$ , where  $g \in [0, 1]$  is the fraction of rationale content words appearing in the evidence and  $h \in [0, 1]$  compares the rationale’s lexical sentiment  $t \in [-1, 1]$  against the score direction  $(\hat{y} - 3)/2$  as  $h = 1 - |t - (\hat{y} - 3)/2|/2$ , with  $h = 0.5$  when the rationale carries no sentiment words.  $A_{\text{std}} = -\max(0, 0.3 - m\Delta) \leq 0$  is a hinge on the score change  $\Delta$  under a rubric perturbed in known direction  $m$ , averaged over the tightened and loosened copies.  $A_{\text{cal}} = |\hat{c}/5 - a|$  with panel agreement  $a = 1 - \min(1, \sigma/2)$ . The item weight is  $w_{\text{conf}} = \bar{c}/5$ . The label-free paraphrase-invariance term is  $A_{\text{para}} = -|\text{score}(S) - \text{score}(\text{paraphrase}(S))|/4$  and enters with weight  $\lambda_{\text{para}}$ . The default preset uses the weights given in Section 4.4. The consistency-based preset raises  $\lambda_s$  to 0.8 and turns on paraphrase invariance at  $\lambda_{\text{para}} = 0.5$ , and varied-standard episodes score the policy toward targets derived from the existing consensus, drawn from perturbed, composite, and contrast standards whose per-video targets are computable from the consensus.

### A.4 STAR-PO TRAINING AND INFERENCE CONFIGURATION

Both stages train a LoRA adapter ( $r = 16$ ,  $\alpha = 32$ , all attention and MLP projections) on Qwen3-8B in 4-bit. Supervised fine-tuning is completion-only with the prompt masked and confidence-weighted,

$$\mathcal{L}_{\text{SFT}} = \frac{\sum_i w_i \text{CE}_i}{\sum_i w_i}, \quad w_i = \frac{c_i}{5}, \quad (8)$$

where  $\text{CE}_i$  is the mean cross-entropy over example  $i$ 's answer tokens, and each example is augmented with a random rubric paraphrase. Preference optimization samples  $K = 6$  candidates per item at temperature 1.0 from the current policy, forms the normalized sign-preserving advantages described in Section 4.4, skips groups whose reward standard deviation is below  $10^{-6}$ , clips gradients at norm 1.0, and applies a per-token KL penalty at  $\beta = 0.15$  to a frozen reference adapter held as a second adapter on the same base, at learning rate  $10^{-4}$  for two epochs. The KL term is the per-token  $k_3$  estimator  $\widehat{D}_{\text{KL}o} = \mathbb{E}_t[\exp(\delta_t) - \delta_t - 1]$  with  $\delta_t = \log \pi_{\text{SFT}} - \log \pi_\theta$  at token  $t$  of candidate  $o$ , so it shares the per-token scale of the log-likelihood term. After each epoch the held-out Spearman correlation is measured, and only a checkpoint that beats the SFT floor is kept. At inference the judge generates one judgment per (video, dimension) and parses the integer score, falling back to the expected value of  $p_\theta$  when parsing fails. Serving runs the judge in its own process behind the arbitration interface of the pipeline.

## A.5 EVIDENCE MEMORY AND RELIABILITY

Cell reliability combines coverage, the amount of valid signal observed, with intrinsic signal confidence. The coverage term is

$$r_{\text{cov}}(n; \tau) = 1 - \exp(-n/\tau), \quad (9)$$

where  $n$  counts the observations supporting the cell and  $\tau$  is the count at which an estimate is treated as well supported. Visual operators use frame counts and speech operators use content-word counts. Prosody reliability is the voiced ratio, because pitch statistics depend on the fraction of frames carrying pitch. Pronunciation reliability is coverage multiplied by ASR confidence. Body delivery reliability is coverage multiplied by keypoint completeness. A missing cell receives  $r = 0$ , which separates a measured value of zero from an absent measurement.

## A.6 COUNCIL THRESHOLDS

A conflict group qualifies when the score spread is at least  $\tau_{\text{gap}} = 1.0$ , the reliability spread is at least  $\tau_{R\text{gap}} = 0.15$ , and at least one member has reliability  $\tau_R = 0.5$ . Reliability-gated consensus pulls each member toward its peers' reliability-weighted mean with a step  $\rho = 0.3$  scaled by its own unreliability. A supplied language model instead proposes a revision bounded to  $\pm 1.0$  per score. Deliberation runs for at most two rounds, and each dimension's uncertainty combines its unreliability with any residual disagreement against its reliable same-modality peers.

## A.7 SPECIALIST AGENT DETAIL

**Threshold-calibrated agents.** Five of the seven agents compare measured features against thresholds derived from the human ratings. For each feature the calibration procedure records the value distribution across the corpus, the feature mean within each human score band, the Spearman correlation between feature and human score, and the direction of the relationship. A presentation is scored per feature by locating its value against the band boundaries. The per-feature scores are combined as a weighted average with weights proportional to the absolute correlation between that feature and the human label. This replaced an earlier rule that started every presentation at 5.0 and subtracted for each violated threshold. That rule concentrated scores in a narrow high band.

**Language agent.** The transcript is reduced to lexical features and a heuristic pre-score. The language model is asked to adjust that pre-score by at most  $\pm 1.0$  given the transcript. The result is clipped to  $[1, 5]$  and the justification is stored in the rationale. Presentations with fewer than 30 transcribed words return a neutral score with the shortfall recorded.

**Slide agents.** The two slide agents read per-slide structures and not the window list. Visual design scores layout, image and chart usage, font variety, and color consistency. Slide content compares the transcript against each slide's text using Jaccard overlap between word sets. This gives three signals, coverage, the fraction of slides whose overlap exceeds a threshold, elaboration, which measures whether the speaker adds material beyond the slide text, and a reference rate counting deictic expressions pointing at the slides. Lexical matching keeps the signals deterministic and inspectable in the rationale.

Table 7: Assessment dimensions and their evidence channels. The judge scores the eight annotated dimensions natively, and the pipeline reports seven channels, since the two *techniques* dimensions share `slide_content`.

| Annotation dimension        | Evidence channel |
|-----------------------------|------------------|
| Language                    | language         |
| Pronunciation (vocal)       | pronunciation    |
| Volume and pace (vocal)     | prosody          |
| Fluency (vocal)             | fluency          |
| Non-vocal delivery          | non_vocal        |
| Slide design (visuals)      | visual_design    |
| Speech–slide sync (visuals) | slide_content    |
| Slide reference (visuals)   | slide_content    |

Table 8: Head-to-head on the overall score, using a paired bootstrap on a shared video resample.  $\Delta\rho$  is AIMPACT minus baseline, and  $p_{\text{Holm}}$  is Holm-Bonferroni adjusted within the dimension. Win rate is the fraction of videos where AIMPACT is closer to the expert.

| Baseline        | $n$ | $\Delta\rho$ | 95% CI          | $p_{\text{Holm}}$ | Win rate |
|-----------------|-----|--------------|-----------------|-------------------|----------|
| Qwen2.5-Omni-7B | 34  | 0.759        | [0.380, 1.094]  | 0.000***          | 0.676    |
| Qwen3-Omni-30B  | 34  | 0.579        | [0.196, 0.956]  | 0.004**           | 0.500    |
| MiniCPM-o-4.5   | 28  | 0.251        | [−0.084, 0.624] | 0.322             | 0.607    |
| Gemini 3.1 Pro  | 34  | 0.153        | [−0.074, 0.400] | 0.322             | 0.382    |

#### A.8 REPORT SYNTHESIS AND FEEDBACK

Issue mining revisits the per-window evidence and flags windows whose metrics cross fixed severity thresholds at three levels, critical, high, and medium. The metrics are filler rate, hesitation rate, longest pause, monotone ratio, low-confidence word counts, low-energy ratio, audience gaze ratio, posture openness, and negative tension. Each flagged window gives a marker recording the time window, dimension, severity, and triggering value. Markers are ordered by severity. Report synthesis lists dimensions scoring at or above 4.0 as strengths and those below 3.0 as weaknesses, reads clarity and energy trends from their signs, and promotes a small set of priority issues where a metric passes a cutoff. A language model turns these into a report with a summary, a per-dimension explanation, time-anchored feedback, and prioritized suggestions. It is instructed to use only the supplied scores, evidence, and insights and not to alter any number. Synthesis is a rendering step and not a scoring one. Visual feedback is computed on shorter windows than scoring, which localizes markers without changing the scores.

#### A.9 DIMENSIONS AND EVIDENCE CHANNELS

The eight annotation dimensions map to seven evidence channels, as summarized in Table 7. The two *techniques* dimensions, speech–slide synchronization and slide reference, share the slide-content evidence channel and are separated only by their rubric text and the sub-metrics that rubric foregrounds. One evidence channel therefore serves two standards under a judge that reads the standard as input.

#### A.10 EXTERNAL-JUDGE HEAD-TO-HEAD AND PER-DIMENSION RANKING

Table 8 gives the head-to-head comparison on the overall score and Table 9 the per-dimension Spearman correlation against the expert, over 34 videos, with MiniCPM reduced to 28 after 6 videos of unparseable output. AIMPACT leads on seven of the eight rows. Gemini 3.1 Pro is nominally higher on prosody. The three open judges turn negative on visual design and slide content, the two dimensions that depend most on reading the slides. Gemini stays positive there at 0.340 and 0.344.

Table 9: Per-dimension Spearman  $\rho$  against the human expert under matched omni inputs, over 34 videos, with MiniCPM at  $n=28$  after 6 videos of unparseable output.

| Dimension     | AIMPACT      | Gemini 3.1 Pro | MiniCPM-o-4.5 | Qwen3-Omni-30B | Qwen2.5-Omni-7B |
|---------------|--------------|----------------|---------------|----------------|-----------------|
| Fluency       | <b>0.766</b> | 0.628          | 0.223         | 0.437          | -0.113          |
| Prosody       | 0.690        | <b>0.695</b>   | 0.405         | 0.280          | 0.152           |
| Pronunciation | <b>0.779</b> | 0.705          | 0.503         | 0.452          | 0.136           |
| Language      | <b>0.665</b> | 0.587          | 0.507         | 0.395          | 0.171           |
| Non-vocal     | <b>0.762</b> | 0.686          | 0.432         | 0.052          | -0.006          |
| Visual design | <b>0.494</b> | 0.340          | 0.074         | -0.281         | -0.197          |
| Slide content | <b>0.487</b> | 0.344          | 0.208         | -0.100         | -0.216          |
| Overall       | <b>0.752</b> | 0.599          | 0.493         | 0.173          | -0.007          |